What Makes a Mission Hard?

Defining, Measuring, and Learning Difficulty in a Game-Scale Gravitational Navigation Environment

Kevin Horecka*

October 30, 2026

Abstract

We present *spacenav*, a simulated navigation environment in which a thrusting spacecraft must visit a sequence of moving targets inside a gravitational field produced by many bodies. The environment is cheap to simulate and easy to state, and yet we find that the ordinary question "how hard is this mission?" resists a single answer in three distinct ways. *First*, difficulty is not one quantity: a mission can be hard because a simple mental model of the gravitational field is wrong, or because the window for acting is narrow, and these two failure modes require opposite fixes. We give a closed-form, simulation-free score, $\mathcal{D} = \eta/A$, that captures the first and can be *inverted* to construct missions at a chosen difficulty, and we argue that it must never be collapsed with the second. *Second*, difficulty is hard to measure even once defined. Our own published conclusion that a class of two-stop missions was infeasible turned out to be an artifact of twelve correlated samples; at thirty-two samples the same class is flown sixteen percent of the time. We report this and two other self-corrections as findings about the measurement problem rather than footnotes. *Third*, difficulty interacts with what the learner is told. Across twenty training runs at five random seeds per condition, we show that adding four hand-designed physical readouts to the agent's observation *costs* it roughly ten percentage points of success, and that the entire cost is attributable to the two readouts that are algebraically redundant given the rest of the observation, while the two that carry genuinely new physics are free. We release the environment, the levels, the trained policies and the analysis code.

1 Introduction

A spacecraft with limited fuel must reach a target that is itself orbiting. The physics is Newtonian and deterministic; the simulation runs in milliseconds; a competent undergraduate can write the equations of motion from memory. And yet if you ask "is this particular mission hard?" the question turns out to have no single well-posed answer, and the ways in which it fails to have one are, we think, more interesting than the environment itself.

This paper is about that failure, using one concrete environment as the subject. *Spacenav* is a two-dimensional, game-scale gravitational navigation task: a ship with a steerable engine and a finite fuel tank must fly through or match velocity with a short sequence of targets, some of which are attached to moving bodies, inside a field generated by up to thirty-two masses. Episodes last 150 simulated seconds at a 15 Hz control rate, and the action space is the nine combinations a

*Correspondence: `kevin.horecka@gmail.com`. Code, levels and every measurement reported here: `https://github.com/kevroy314/spaceship-navigator`.

human could press on a gamepad. The environment is designed as a testbed for reinforcement learning, but the difficulties we report are not specific to learning: a search-based trajectory optimiser meets most of them too.

We make three claims.

Difficulty needs at least two axes, and one of them has a closed form. The natural mesoscale model of a gravitational field — "only the body I am orbiting matters", known in astrodynamics as *patched conics* — is wrong by a measurable amount at every point in space. We define η as the fraction of the local gravitational acceleration that such a one-body model discards, and A as the ship's fuel budget expressed in units of the impulse gravity will deliver over the remaining flight. Their ratio $\mathcal{D} = \eta/A$ is the share of the tank that flying the wrong model would cost. It requires no simulation, so millions of candidate missions can be scored for free, and it inverts: given a target difficulty one can solve for the fuel that places a mission there. But $\mathcal{D}$ says nothing about *timing*, and a mission can be trivial to plan and impossible to execute. We keep the two separate throughout, and show what collapsing them would hide (Section 4).

Measuring difficulty is itself hard, and we got it wrong in public. The honest way to ask whether a mission class is feasible is to let a strong optimiser attempt it. We did that, with three search budgets, and reported that a class of two-stop missions never succeeded. The result was wrong: the three budgets had been run against the *same* twelve missions, so they were not three independent attempts, and the true success rate is about sixteen percent, at which drawing zero out of twelve has probability 0.13. We describe this and two further self-corrections in Section 6, because a benchmark whose difficulty estimates are this fragile is a fact about the benchmark.

What you tell the agent changes what it can learn, in a direction that is easy to get backwards. We instrumented the environment with four physically meaningful readouts — η, the fuel-versus-gravity ratio, their quotient $\mathcal{D}$, and a local tidal stretching rate — on the hypothesis that an agent told about the structure of its own difficulty would learn faster. The opposite happened. Across five training seeds per condition, the four channels cost 3.1 to 10.1 percentage points of final success depending on the evaluation tier, all four differences significant. Splitting the panel in half shows the cost is *entirely* carried by the two channels that are algebraic combinations of quantities the agent already observes; the two that carry information not otherwise present are statistically indistinguishable from supplying nothing at all (Section 8).

Throughout, we have tried to report what we measured rather than what we expected, including where the two differed. The environment's own development history is littered with elegant, confident and wrong conclusions, each caught only by a measurement, and we have found that record more useful than the successes.

2 Related work

Astrodynamics and the coarse model. The idea that one body dominates a spacecraft's local dynamics is the *patched-conic approximation*, standard in mission design and accurate precisely when the quantity we call η is small. Its domain of validity is usually described by the Hill radius or the Laplace sphere of influence. We instead use the *equal-pull surface*, the radius $d = a\sqrt{m/M}$ at which a secondary of mass m at semi-major axis a pulls as hard as its primary of mass M; this is roughly half a Hill radius and is where η peaks. Learning-based guidance for

low-thrust transfers has been studied by Izzo and Öztürk [2021] among others, who report that policy networks trained on optimal transfers recover near-optimal control with small optimality gaps; our setting differs in being game-scale, partially observed and adversarially generated rather than drawn from a mission catalogue.

Chaos, and whether it is present. Our initial design intent was for the environment to be chaotic, since chaotic regimes are where coarse mental models fail most interestingly. The standard route to constructed chaos in a planetary context is the Chirikov resonance overlap criterion [Chirikov, 1979] and Wisdom's 2/7 law for the chaotic zone around a secondary [Wisdom, 1980], refined by Deck et al. [2013]. We rejected this route on timescale grounds and verified the rejection empirically (Section 5): resonance overlap governs evolution over *many* orbital periods, whereas an episode here is a fraction of one. For basin structure we use the uncertainty exponent α of Grebogi et al. [1983], under which the fraction of uncertain initial conditions scales as $f(\varepsilon) \sim \varepsilon^{\alpha}$, with $\alpha \approx 1$ indicating a smooth, resolvable boundary and $\alpha \approx 0$ a fractal one; basin entropy [Daza et al., 2016] is a related diagnostic.

Why chaos would matter for a learner. Wang et al. [2023] show that a reinforcement learning objective is α-Hölder with $\alpha = \log(1/\gamma)/\lambda$ for a system with maximal Lyapunov exponent λ and discount γ, and becomes fractal — losing any useful descent direction — when λ exceeds $\log(1/\gamma)/\Delta t$. At our discount of $\gamma = 0.999$ and control interval $\Delta t = 1/15\,\mathrm{s}$ this threshold is $\lambda_{\mathrm{crit}} = 0.0150\,\mathrm{s}^{-1}$. We measured against it rather than assuming either side of it.

Policy compression and execution precision. That "how simple is the rule" and "how precisely must it be timed" are different quantities is argued from two directions: policy compression in human decision making [Lai and Gershman, 2024] and policy information capacity as a task-difficulty measure [Furuta et al., 2021]. A bang-bang controller on a razor-thin switching surface is a one-bit policy on an impossible trigger. Our execution-margin thresholds are taken from the human motor literature: approximately 20 ms onset jitter for a practised discrete action [Repp, 2005] and a closed-loop correction ceiling near 3 Hz [Franklin et al., 2019].

Evaluation practice. Our statistical protocol follows Agarwal et al. [2021] and Colas et al. [2018]: separate training-seed variance from episode variance, use enough seeds to say anything about the former, and prefer interval estimates. This discipline is the reason several results in this paper are reported as null. For context on the scale of modern environment-step budgets we cite Matthews et al. [2024] and Lee et al. [2025], both of which make clear that our training budget is small by current standards — a limitation we return to in Section 12.

3 The environment

3.1 Physics

The simulation is two-dimensional with gravitational constant $G = 1$, so each "mass" is really a gravitational parameter GM. Lengths are game units (u) and times are game seconds (s); a typical level spans 1000–3000 u. Bodies attract one another and are integrated with a kick-drift-kick leapfrog scheme at a substep of 1/60 s. Within a body's radius the field falls off linearly, as for a uniform sphere, which keeps the acceleration finite everywhere and makes self-interaction exactly zero.

The ship is a *test particle*: it feels every body but moves none. This is the single most consequential design decision in the environment. Because the ship exerts no force, a level's body trajectories are completely determined by its initial snapshot, which means (i) the entire future of the field can be precomputed and replayed exactly, (ii) a browser client can reproduce the physics from a $\sim$1.5 KB snapshot by integrating the bodies itself, and (iii) a level is reproducible from a seed with no stored trajectory data.

The ship's acceleration is the sum of gravity, atmospheric drag where a body has an atmosphere, and thrust. Thrust is held constant across a control tick. Four physics substeps make one control tick, so the agent and a human player both act at 15 Hz, or every 66.7 ms.

3.2 Missions, hazards and termination

A *mission* is a short tour: between one and six targets, each either a point in free space, a station, or a point attached to a moving body. Each target carries an arrival radius (default 12 u) and a velocity tolerance; a flythrough target accepts any speed, while a *rendezvous* target requires matching to within 3 u/s, which is a materially harder manoeuvre. A tour may be ordered or order-free. The ship carries a finite tank, expressed as seconds of full-throttle burn, and a per-mission engine acceleration; both are sampled per mission from ranges chosen so that a ship can never simply ignore gravity and fly straight at the target.

Hazards are radiation from luminous bodies, atmospheric heating, debris fields and radiation belts. An episode ends on arrival at the final target, collision with a solid body, destruction by accumulated damage, fuel exhaustion with no remaining way to arrive, departure past 1.35 times the level radius, or timeout at 150 s (2250 control ticks).

3.3 Observation and action

The observation is ego-centric and set-structured, so that the same weights run on a level with any number of bodies. It comprises a fixed ship-and-tour block (velocity, local gravity vector, relative target position and velocity, fuel fraction, remaining Δv against the tour's estimated requirement, health, time remaining, and the objective weights), one token per body (relative position and velocity, distance, the body's contribution to local acceleration, mass, radius, atmosphere, luminosity, and a one-hot kind), and one token per hazard zone. Bodies and zones are embedded token-wise and pooled two ways: multi-head cross-attention with the ship's own features as the query, and a masked max-pool.

The action space is deliberately the set a person can press: turn $\in \{\text{left}, \text{none}, \text{right}\}$ crossed with throttle $\in \{\text{off}, \text{gentle}, \text{full}\}$, nine discrete actions in total. Turning is arcade-style with no angular inertia.

3.4 Level families and pools

Levels are generated procedurally in twelve named families, from `sol_like` (a dominant star with well-separated planets) through `binary_star`, `trinary`, `star_cluster`, `bh_cluster`, `debris_maze` and `rogue_flyby`. A generated level is accepted only if it remains well-behaved — no collisions, no ejections, bounded energy drift — for 900 s, six times a single episode. Accepted levels are stored as snapshots at 20 s intervals so that an episode may begin at any of several epochs in the level's evolution.

Four pools are used: a training pool of 18,000 levels, a validation pool of 2,700 levels drawn from the same families, a holdout pool of 1,200 levels from families never seen in training, and a

long-horizon pool of 1,800 levels carrying sixteen snapshots for experiments at extended episode clocks. Pools are reproducible from seeds.

4 Defining difficulty

4.1 The coarse model and what it discards

The mental model a person brings to a gravitational field is almost always the one-body model: identify the dominant attractor, treat its field as the whole field, and reason about conic sections. We instantiate exactly this model in code. The function `gravity_at(..., topk=k)` restricts the ship's felt gravity to the k strongest attractors at its current position and discards the rest; $k = 1$ is patched conics, and $k = 0$ is the truth. Bodies always move truthfully — only the test particle's felt force is restricted — so a plan made inside the coarse model can be flown in the true world and the discrepancy observed directly.

This gives the first axis of difficulty a natural definition. Let a_{full} be the true local acceleration and a_{dom} the acceleration from the single strongest body. Define

$$\eta \;=\; \frac{\|a_{\text{full}} - a_{\text{dom}}\|}{\|a_{\text{full}}\|}, \tag{1}$$

the fraction of the local field that a one-body picture throws away. Near zero, patched conics is right; near one, it is lying.

4.2 Authority, and the ratio that matters

η alone is not difficulty, because a ship with a large enough engine can simply absorb the error. Let Δv_{tank} be the mission's fuel expressed as a velocity budget, $\|g\|$ the local gravitational acceleration and T the remaining flight time. Define the *authority*

$$A \;=\; \frac{\Delta v_{\text{tank}}}{\|g\|\,T}, \tag{2}$$

the tank measured in units of the impulse gravity itself will deliver over the flight. $A > 1$ means the ship can out-push the field; $A < 1$ means it cannot.

The quantity we use to score a mission is the ratio

$$\boxed{\mathcal{D} \;=\; \frac{\eta}{A}} \tag{3}$$

which has a direct reading: *the share of the fuel tank that flying the wrong model would cost.* Both ingredients are available in closed form from the level snapshot and the mission specification, so $\mathcal{D}$ requires no simulation at all; thousands of candidate missions can be scored essentially for free, and expensive planning is spent only on missions already known to sit where they are wanted.

$\mathcal{D}$ also *inverts*. Solving for the tank,

$$\Delta v_{\text{tank}} \;=\; \frac{\eta\,\|g\|\,T}{\mathcal{D}}, \tag{4}$$

so a mission can be placed on a chosen difficulty rung by solving rather than by generate-and-test. One practical caution: when retuning a mission onto a rung we only ever *add* fuel. Reaching a high rung by starving the tank produces an infeasible mission dressed up as a demanding one, which is a different thing.

Calibration. An ordinary mission in this environment sits at $\mathcal{D} \approx 0.0014$, with $\eta = 0.0036$ and $A = 3.8$. Gravity supplies about a quarter of the tank's worth of impulse over a flight, but a *single* body explains 99.6% of the field. That is why patched conics works here, and why breaking it requires deliberate construction rather than luck.

Validation. We compared a planner restricted to the one-body model against a planner with the true field, on missions pooled by $\mathcal{D}$ band, 24 missions per band. Below $\mathcal{D} = 0.1$ the two tie (32% versus 42% arrival, Fisher exact $p = 0.74$); above it they separate sharply (13% versus 54%, $p = 0.005$), with the coarse planner failing every mission on the top rung. We note explicitly that the *per-rung* figures, at $n = 8$, are not significant; only the pooled comparison is. This is the first of several places where our own early reporting outran its sample size.

4.3 Where to put a target: the equal-pull surface

To raise η constructively one needs a place to stand. The literature's obvious candidate is the chaotic zone around a secondary, delimited by the resonance-overlap criterion. We tried it and rejected it. Resonance overlap describes what happens over many orbital periods; an episode here is a fraction of one, so asymptotic chaos never gets time to act. The measurement was blunt: placing targets inside the Wisdom band lifted η from 0.0026 to 0.0172, a sixfold increase that is still negligible, because a few Hill radii from a secondary is an $\eta \approx 0.02$ desert.

What governs an episode this short is instantaneous force competition. The relevant surface is where a secondary's pull equals its primary's, at $d = a\sqrt{m/M}$ — about half a Hill radius. Figure 1(a) shows η measured on a shell around the second-most-massive body of 297 levels spanning nine families, taken as a median over 24 directions at each radius. It rises from 0.09 at 0.3× the equal-pull radius to a peak of 0.43 at 1.0×, then falls to 0.24, 0.066 and 0.020 at 2×, 4× and 8×. Target placement in the contested-mission generator is therefore confined to the 0.8–1.3× shell.

We note that the peak value depends strongly on how one samples it. Along the line joining primary and secondary the two pulls oppose, the denominator nearly cancels and η exceeds 1; the angular median reported here is the conservative statistic and is the one the generator is tuned against.

The larger lever is family choice (Figure 1(b)). Median η over the whole playable area is 0.0005 for `debris_maze` and 0.0027 for `sol_like`, against 0.35 for `binary_star` and 0.53 for `star_cluster` — a spread of three orders of magnitude. Systems of comparable masses are contested everywhere; hierarchical systems are contested only in thin shells that must be aimed at deliberately.

4.4 The second axis, and why it must stay separate

$\mathcal{D}$ measures whether a coarse *model* suffices. It says nothing about whether a correct plan can be *executed*. We therefore band every mission on two axes and never reduce them to one number:

- **Model scale** — the coarsest k in `gravity_at(topk=k)` under which a planner's trajectory still arrives when flown in the true world.
- **Execution margin** — timing slack against measured human limits: ∼20 ms motor onset jitter, ∼200 ms visual reaction time, and a 3 Hz closed-loop correction ceiling. Our control tick is 66.7 ms, already coarser than a trained hand, which makes the comparison generous rather than rigged.

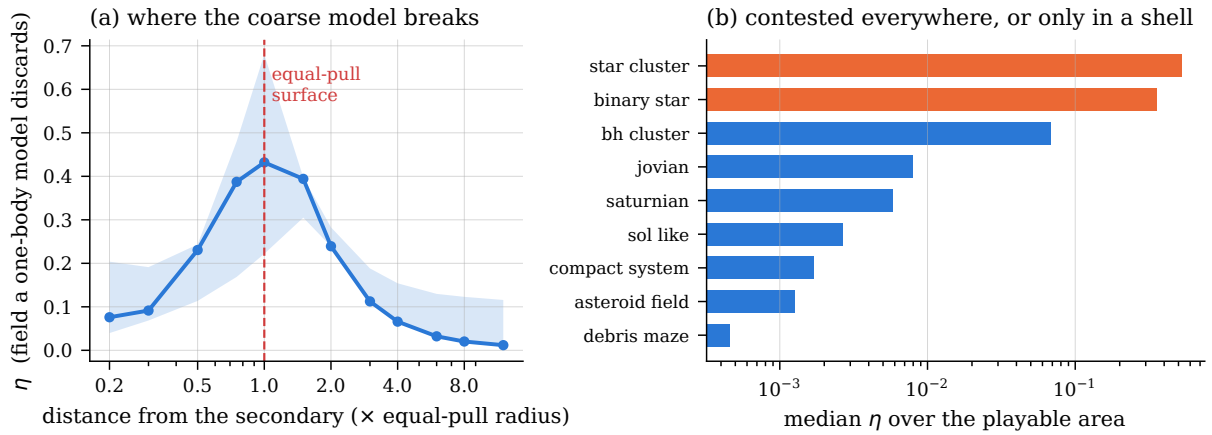


Figure 1: (a) The fraction η of local gravity that a one-body model discards, against distance from a secondary in units of the equal-pull radius $a\sqrt{m/M}$. Median over 24 directions on each shell, across 297 levels from nine families; the band is the interquartile range. The coarse model is worst almost exactly where the two pulls balance. (b) Median η over the whole playable area, by level family. Families in orange are contested everywhere; the rest are contested only inside a thin shell.

The resulting bands (Figure 2) are *trivial*, *human*, *calculated* (intuition fails but computation succeeds), *precision* (no hand can hold it) and *opaque* (nothing we have succeeds; rejected as unfair). The reason to keep the axes apart is that *calculated* and *precision* are different kinds of impossible and need opposite repairs: widen a *precision* mission's timing window and it becomes *calculated*; simplify a *calculated* mission's field and it becomes *human*. A single scalar difficulty would merge exactly the two cases a level designer most needs to tell apart.

5 Is the environment chaotic?

Several of our design threads — fractal basin boundaries, a "lottery" difficulty band, and a worry that a fractal objective would leave a policy gradient with no descent direction — presupposed that the field was chaotic on episode timescales. It is not, and establishing that changed the plan.

The concern with sharpest teeth was the Hölder result of Wang et al. [2023]: at $\gamma = 0.999$ and 15 Hz, a maximal Lyapunov exponent above $\lambda_{\text{crit}} = 0.0150\,\text{s}^{-1}$ would make the objective fractal. Our measured predictability horizon — the time at which one control tick's worth of thrust error grows past the 12 u arrival tolerance — is 52–74 s across eight families, uncomfortably close to the 67 s corresponding to that threshold.

The resolution is that the divergence is *ballistic, not exponential*. The discriminator is the decay of $d\ln(\text{sep})/dt$, which falls as $1/t$ for linear growth and stays flat for exponential growth; pure linear growth predicts a 150 s to 75 s slope ratio of exactly 0.50, and we measure 0.33–0.64 with median ≈ 0.56. Fitting $\text{sep} = a\,t\,e^{\lambda t}$ to divide out the unavoidable ballistic factor gives $\lambda = 0.0045$–$0.0086\,\text{s}^{-1}$ (90th percentile ≈ 0.010), roughly a factor of two below λ_{crit}; only the tail of `star_cluster` exceeds it. A Lyapunov time of ~ 140 s against 150 s episodes is about one e-folding per episode, or ~ 2000 control steps per e-folding, whereas the successful chaotic-control results in the literature sit at 10–100 steps per e-folding. We are comfortably inside the benign regime.

The uncertainty exponent told the same story the hard way. Our first estimates of α were

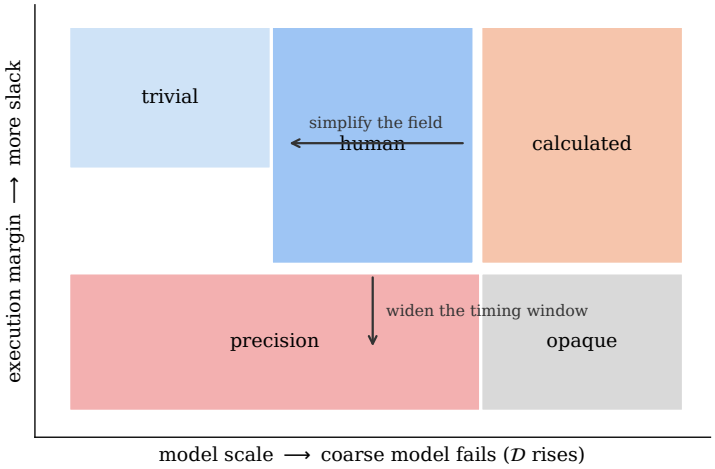


Figure 2: The two difficulty axes kept apart. A mission's position on the horizontal axis says whether a coarse one-body model suffices; its position on the vertical axis says whether the timing window is within human reach. *Calculated* and *precision* are different kinds of impossible and need opposite repairs, which is why a single scalar difficulty would merge exactly the two cases a designer most needs to tell apart.

computed from too few samples: with 96 perturbation samples the boundary-flip counts were small integers (0, 1, 3, 5, 5), one fitted exponent of 1.297 was a two-point slope between two and six flips, and bootstrapped confidence intervals straddled the threshold separating fractal from resolvable. Zero of five exponents was trustworthy, and we had reported a fractal/resolvable split that the data did not support. After raising the sample count eightfold, α measures 0.53–1.02: resolvable, consistent with the Lyapunov result. We also added an explicit `measurable` flag so that an unmeasurable exponent propagates as *unknown* rather than as *fractal*.

Three loose ends resolve at once. The environment is *quasi-integrable* on episode timescales. Two difficulty bands that were supposed to capture chaotic missions contain zero instances, because there is no chaos to capture. And a previously published claim of ours, that a training plateau was caused by a fractal objective, is retracted: that plateau needs a different explanation.

6 Measuring difficulty empirically

6.1 The planner as yardstick, and the solvability gate

Closed-form $\mathcal{D}$ scores a mission's model demand, but it cannot tell you whether a mission is flyable at all. For that we use a cross-entropy-method trajectory optimiser as a reference pilot: piecewise-constant control segments over the episode, seeded from a scripted reflex controller and from an explicit zero-thrust candidate, iteratively refined toward an arrival objective. A mission counts as *solvable* if the optimiser arrives, or if any member of an enumerated one- or two-burn family arrives. Missions that nothing can fly are excluded from difficulty labelling: a level nobody can fly earns no label, since its measured difficulty would describe the probe rather than the level.

Including the zero-thrust candidate matters more than it sounds. Throttle is parameterised through a sigmoid, so an all-zero-throttle plan is essentially never drawn by the search; on missions whose optimal solution is to coast and let gravity do the work, the optimiser was therefore failing on the easiest levels in the environment until the candidate was added explicitly.

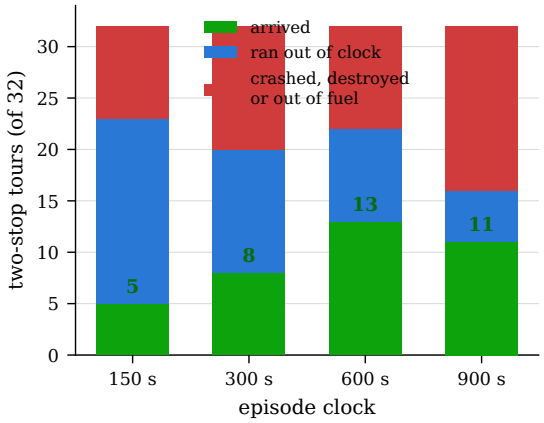


Figure 3: Thirty-two two-stop missions flown by the reference optimiser at four episode clocks, with control resolution held at 3.1 s per segment so that a longer clock buys more segments rather than coarser ones. Timeouts fall monotonically; arrivals peak near 600 s; hazard endings rise throughout. An earlier report of ours, based on twelve correlated samples at the 150 s clock, concluded that this mission class was infeasible.

6.2 A retraction, and what it says about the benchmark

We previously reported that two-stop missions — tours with two sequential targets, typically with at least one rendezvous — were *infeasible* at the 150 s episode clock rather than merely hard. The evidence was that twelve such missions were attempted under three search budgets, including one with sixteen times the trajectory evaluations, and none succeeded, with timeout as the dominant failure.

That conclusion was wrong, and the way it was wrong is instructive. The three budgets were run against the *same* twelve missions, so they were close to perfectly correlated and did not constitute three independent chances to observe a success. Re-running with thirty-two missions per condition and a modestly larger search gives a success rate of $5/32 \approx 16\%$ at the same clock. At that rate, observing zero successes in twelve attempts has probability $(1 - 5/32)^{12} = 0.13$ — an unremarkable outcome, not evidence of impossibility.

Extending the episode clock while holding control resolution fixed at 3.1 s per segment gives the picture in Figure 3. Arrivals rise from 5/32 at 150 s to 13/32 at 600 s and then stop; timeouts fall monotonically from 18 to 5; and the episodes that a longer clock frees up become hazard endings about as often as they become arrivals: crashes, destructions and fuel exhaustions go from 9 at the shortest clock to 16 at the longest, though not monotonically (9, 12, 10, 16). More time buys more exposure. The useful ceiling for this mission class is around 600 s, and the binding constraint beyond it is hazard exposure rather than the clock.

6.3 What the episode clock is, structurally

The episode clock is a global constant compared against a tick counter, not a per-mission field. This is a limitation we discovered while running the sweep above — each clock required a separate process with the constant patched at import — and it is the reason the environment cannot presently run long missions for play while measuring difficulty on a short, fixed clock. We note it because it is exactly the kind of structural detail that silently constrains which experiments are cheap.

	Easy	Medium	Hard	Blind
Learned policy (5 seeds)	89.4 ± 1.5	41.7 ± 3.0	10.7 ± 1.2	30.2 ± 2.3
Scripted reflex pilot	85.3	34.7	7.6	31.8
Difference	+4.1	+7.0	+3.1	**−1.6**

Table 1: Percentage of missions arrived. The learned policy beats the scripted pilot on three suites and loses on *blind*, the only suite drawn from families never seen in training and the one where tour order is sometimes the pilot's to choose.

7 Learning

7.1 Agent and training

We train with Proximal Policy Optimization [Schulman et al., 2017] in a fully vectorised setup where rollout, advantage estimation and all minibatch epochs compile into a single accelerated call. The policy is the actor-critic described in Section 3: embedding width 128, four attention heads, 285,322 parameters in total.

Hyperparameters: 1024 parallel environments × 128 steps = 131,072 transitions per update, four epochs, eight minibatches, learning rate 3×10^{-4} annealed linearly to a tenth of that, $\gamma = 0.999$, generalised-advantage $\lambda = 0.95$, clip 0.2, entropy coefficient annealed from 0.01 to 0.002, value coefficient 0.5, gradient-norm clip 0.5. Runs are 3000 updates, or 393 million environment steps.

A curriculum ramps three things over training: the probability that a leg is a rendezvous rather than a flythrough ($0.1 \to 0.45$), the maximum tour length ($1 \to 3$), and the proportion of missions whose targets are placed in the contested equal-pull shell ($0 \to 0.35$, held at zero until the ship can fly at all). The fuel budget and engine acceleration start generous and tighten to their full ranges.

7.2 Evaluation suites

Four frozen suites of approximately 600 missions each are held fixed across all runs: *easy* (one flythrough target, roomy tank, strong engine, three hierarchical families), *medium* (one or two targets, some rendezvous, tighter budget, all training families), *hard* (two or three targets, half of them rendezvous, a budget usually insufficient for a direct flight so that gravity must do some of the work), and *blind* (families never seen in training, with order-free tours 30% of the time). Six hundred missions per suite gives roughly two percentage points of binomial noise on a success rate.

Each suite also carries a reference score from a scripted reflex pilot — proportional navigation with gravity compensation — evaluated on the identical missions.

7.3 Results

Table 1 gives final performance for the best configuration against the reflex pilot. Success is the mean of each run's last three checkpoints, averaged over five training seeds.

Two observations deserve emphasis. First, the learned policy *loses* to a hand-written reflex controller on the held-out families. Second, the reflex pilot is a weaker yardstick than it appears: on a comfortable-tank mission set it arrives 63.3% of the time, and its dominant failure is *collision* at 21.6%, not fuel exhaustion (4.7%) or timeout (7.8%). Any margin quoted against this opponent

	Easy	Medium	Hard	Blind
No channels	89.4 ± 1.50	41.7 ± 3.04	10.7 ± 1.24	30.2 ± 2.27
η, stretch	90.4 ± 1.81	41.9 ± 2.30	11.1 ± 1.03	29.2 ± 1.23
budget, demand	86.6 ± 1.22	33.2 ± 2.61	6.2 ± 1.68	22.7 ± 2.52
All four	86.3 ± 1.87	31.6 ± 5.25	5.8 ± 3.39	22.2 ± 2.99
p (none vs. η,stretch)	0.40	0.90	0.60	0.42
p (all vs. budget,demand)	0.77	0.59	0.81	0.79
p (η,stretch vs. budget,demand)	**0.0064**	**0.0005**	**0.0011**	**0.0022**

Table 2: Percentage of missions arrived, mean ± standard deviation over five training seeds per arm, 3000 updates each. Welch's t-test on seed means. The two informative channels are statistically indistinguishable from supplying nothing; the two redundant channels are indistinguishable from supplying all four.

is correspondingly soft, a caveat we attach to every such comparison. Absolute success rates are unaffected.

8 What the agent should be told

8.1 Motivation and design

Having defined η, A and $\mathcal{D}$ as the quantities that describe a mission's difficulty, the natural question is whether an agent told those quantities learns better. We added four channels to the observation: η; a log-compressed authority A; a log-compressed demand $\mathcal{D}$; and a local tidal stretching rate, the square root of the tidal gradient, which measures how fast nearby trajectories are pulled apart. These are intended as a cockpit: the quantities a well-instrumented pilot would read.

The channels can be zeroed individually or together, leaving the observation *shape* unchanged, so that an ablated run remains comparable with a full one and no checkpoint is invalidated. We verified separately that toggling them does not perturb the dynamics.

8.2 The four channels cost roughly ten points

Figure 4(a) and Table 2 give the result at five training seeds per arm. Supplying all four channels costs 3.1 to 10.1 percentage points of final success, with Welch's t-test on seed means significant on all four suites and effect sizes between 1.8 and 3.0. We emphasise that these are tests on *seed means*: the quoted uncertainties are training-seed standard deviations, not episode-level binomial intervals, which are much smaller and would overstate confidence.

8.3 The cost is carried entirely by the redundant channels

The explanation was visible in the observation code before it was measured. The observation already carries the local gravity vector (hence $\|g\|$), the product of fuel and engine acceleration (hence Δv_{tank}), and the fraction of the episode remaining (hence T). Therefore $A = \Delta v_{\text{tank}}/(\|g\|T)$ is an algebraic combination of three quantities the agent already has, and partly duplicates an existing ratio of remaining to required Δv; and $\mathcal{D} = \eta/A$ is redundant given η and A. By contrast η requires the dominant-body decomposition, which is not otherwise computable from the observation, and the tidal stretching rate is not present at all.

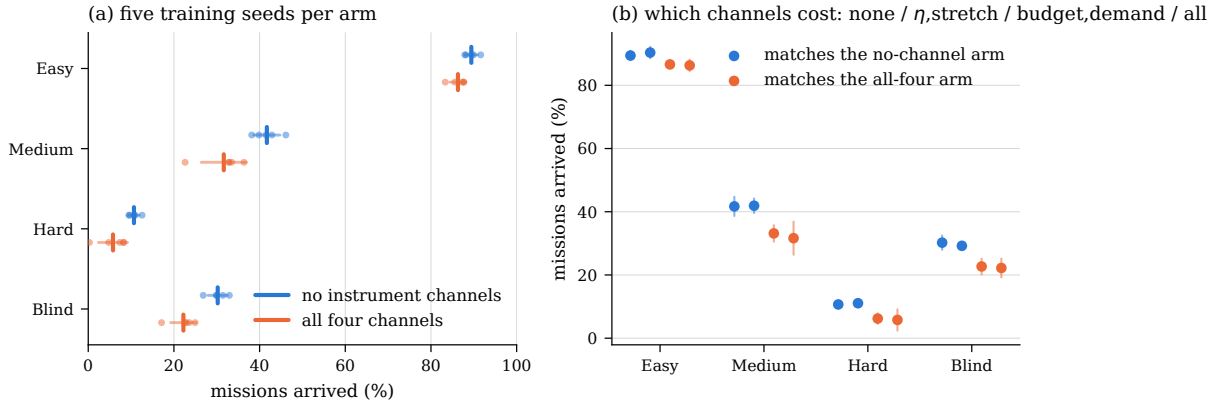


Figure 4: (a) Final success for every training seed, with no instrument channels against all four. Dots are individual seeds, bars are one standard deviation, the heavy tick is the mean. (b) The four-way split. In every suite the no-channel and η,stretch arms coincide, and the budget,demand and all-four arms coincide below them.

This generated a prediction: splitting the panel along the redundancy line should put the entire cost on the redundant half. It does. Supplying only η and the stretching rate is statistically indistinguishable from supplying nothing ($p = 0.40$ to 0.90 across suites); supplying only A and $\mathcal{D}$ is indistinguishable from supplying all four ($p = 0.59$ to 0.81); and the contrast between the two halves is significant on every suite with effect sizes from 2.4 to 3.6.

We draw a narrow conclusion and resist a broad one. Narrowly: in this environment, at this network size and step budget, appending algebraically redundant features to an observation measurably degrades a policy, while appending features carrying genuinely new physics is free. We do *not* have evidence that the informative channels help — η and stretch are nominally above the no-channel arm on three suites and below on one, none significantly. And we have not isolated the mechanism. Four extra input dimensions against 285,322 parameters is a capacity story; a ratio with a division in it is heavy-tailed even after log compression, which is a conditioning story. Both remain untested.

9 What a single number hides

Evaluation reports one success rate per checkpoint, computed from a greedy policy that takes the most probable action. That number conceals both the policy's own stochasticity and the enormous variation between individual missions. Figure 5 makes both visible: every checkpoint of one training run flying one fixed mission sixteen times, with actions *sampled* rather than maximised, for $31 \times 16 = 496$ trajectories per mission.

Two things follow. First, sampled success is far below greedy success — the *easy* mission shown arrives 52% of the time when sampled against 89% for the greedy policy on that suite — so the headline number describes a behaviour the policy exhibits only when its stochasticity is removed. Second, the spread *between missions within a suite* is larger than any effect we measure between training configurations: 60%, 52%, 8% and 0% on the four missions shown. The instrument-channel effect of Section 8 is seven to ten points; two missions drawn from the same suite differ by fifty. Any claim about this environment that rests on a handful of missions is therefore fragile by construction, which is the lesson of Section 6.2 restated in pictures.

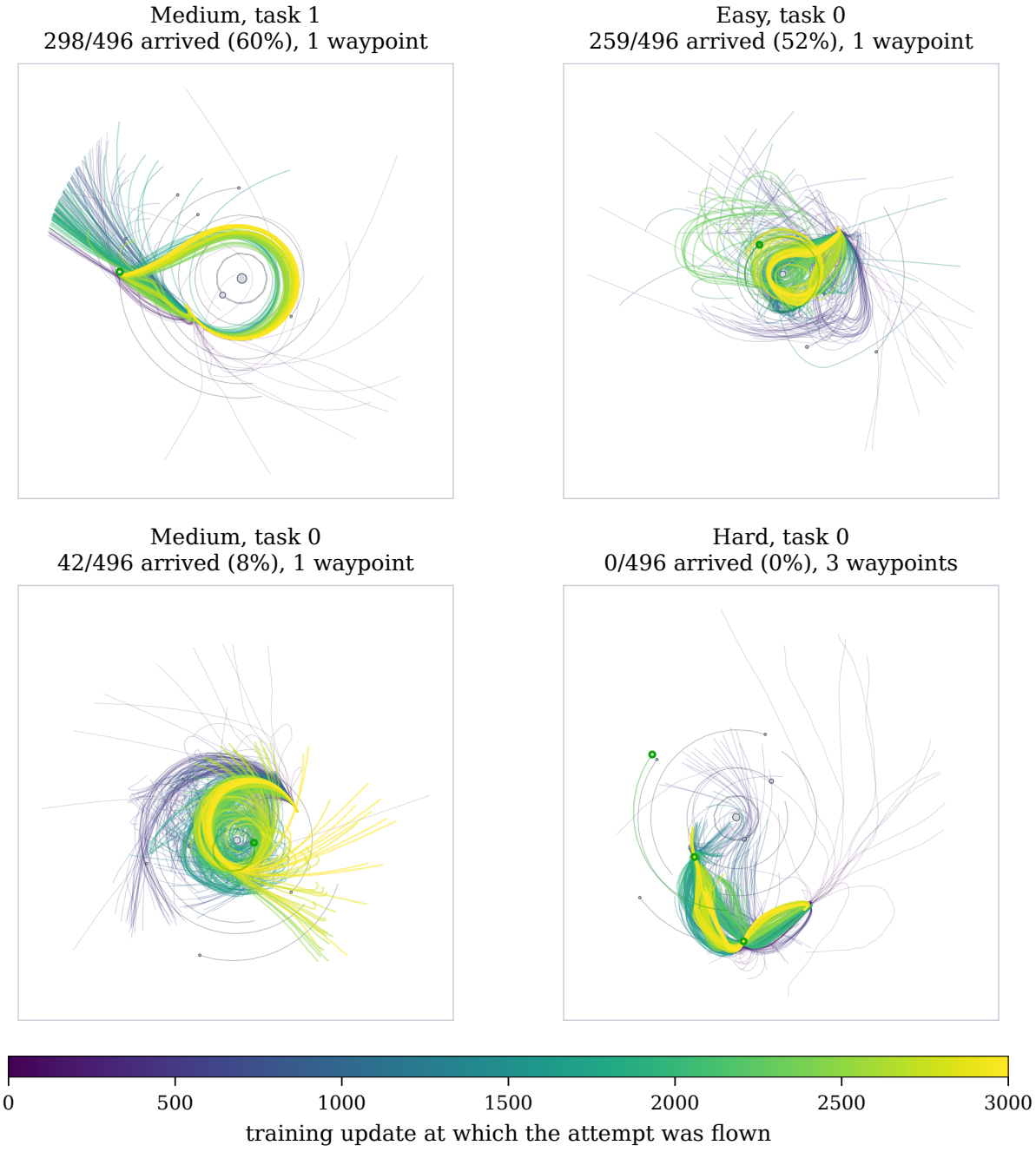


Figure 5: Four missions from the evaluation suites, each flown 496 times by the same training run: every saved checkpoint, sixteen sampled rollouts each, coloured by the training update at which the attempt was flown. Grey discs are bodies with their orbits traced faintly; green rings are targets, drawn with their own paths where they are attached to moving bodies. Early policies (dark) spray across the field; late policies (bright) converge into tight bundles. *Medium, task 1* converges onto a single arrival corridor. *Medium, task 0* converges onto a corridor that mostly misses. *Hard, task 0* converges onto a tight shuttle between its first two targets and never reaches the third, which sits at upper left: an agent confidently executing a plan that cannot complete the mission.

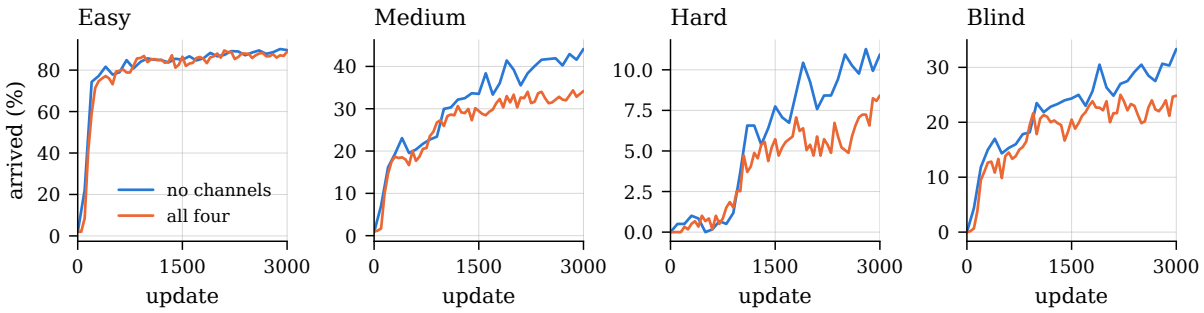


Figure 6: Evaluation success against training update for one seed of each arm. Mid-run checkpoints are noisy enough that reading an ablation before it finishes is actively misleading; we did so twice during this work and drew the wrong conclusion both times.

Figure 6 shows the corresponding learning curves.

10 Threats to validity

The reference pilot is weak. Every margin quoted against the scripted pilot is softened by its 21.6% collision rate (Section 7). Absolute rates are unaffected.

The hard suite is partly unflyable. The reference optimiser arrives on only about 16% of two-stop missions at the measurement clock, and the *hard* suite draws from that family. An agent scoring 10.7% against a 3000-trajectory-per-decision optimiser scoring 16% is a different statement from an agent scoring 10.7% against a solvable task.

Training budget. At 393 million environment steps our runs are small relative to contemporary work at the 10^{11} scale [Matthews et al., 2024], and at 285,322 parameters the network is small relative to current general-purpose agents [Lee et al., 2025]. Results here should be read as statements about this operating point, not asymptotically.

Seeds. Five training seeds per arm meets the minimum we set ourselves, following Agarwal et al. [2021], but it is a minimum. Interval estimates on effect sizes at $n = 5$ remain wide.

Two-dimensional physics. The environment is planar. Several phenomena that matter in real mission design — plane changes, out-of-plane perturbations — cannot occur.

11 Reproduction

All code, level generators, evaluation suites and analysis scripts are at `https://github.com/kevroy314/spaceship-navigator`. The environment is implemented in JAX; training runs reported here were executed on a single consumer GPU. The repository additionally contains a numbered, append-only series of decision records documenting the reasoning behind each design choice, including the ones that were later retracted; most of the self-corrections reported in this paper are traceable to a specific record and the measurement that forced it.

Representative commands:

```
python scripts/build_pools.py                  # regenerate level pools from seeds
python scripts/train.py --run myrun --updates 3000 --seed 0
SPACENAV_INSTRUMENTS=eta,stretch python scripts/train.py --run instr --updates 3000
python scripts/twostop_clock.py --clock 600 --per-spec 8 --out clock_600.json
python scripts/ghost_fleet.py --run myrun --suite medium --task 1 --rollouts 16
python scripts/paper_figures.py
```

The instrument channels are selected by the `SPACENAV_INSTRUMENTS` environment variable, which accepts a named subset, a four-bit mask, or `0/1`. A checkpoint must be evaluated with the same setting it was trained with; the observation shape is identical either way, so a mismatch produces a policy reading zeros where it expects an instrument, with no error.

12 Limitations and outlook

The most substantive limitation is that we do not know *why* redundant observation features cost a policy ten points. Two mechanisms — capacity pressure and the conditioning of heavy-tailed ratio features — predict different behaviour under scale, and distinguishing them requires training at embedding widths we have not been able to afford at five seeds per arm.

Second, the environment is not chaotic, and we would like it to be. Section 5 suggests the route: closer encounters, tighter binaries, and above all episodes that are long relative to orbital periods. That requires a per-mission episode clock, which the current implementation does not have.

Third, our difficulty labels are only as good as the reference optimiser. A mission labelled *opaque* is one nothing we have tried can fly, which is a statement about our search as much as about the mission.

What we think transfers beyond this particular environment is the shape of the problem. A task can be cheap to simulate, deterministic, fully specified, and still resist a one-dimensional notion of difficulty; the measurement of that difficulty can be fragile enough that a published conclusion inverts on a larger sample; and the information given to a learner about its own task can hurt it in ways that depend not on whether the information is *true* but on whether it is *new*. None of those three is specific to gravity.

Acknowledgements

Development of this environment, including the experimental design, measurement code and analysis reported here, was carried out in collaboration with Claude (Anthropic), operating as an autonomous agent with access to the repository and compute. The decision records referenced in Section 11 were written during that process and include the agent's own retracted conclusions.

References

R. Agarwal, M. Schwarzer, P. S. Castro, A. Courville, and M. G. Bellemare. Deep reinforcement learning at the edge of the statistical precipice. *Advances in Neural Information Processing Systems*, 2021. arXiv:2108.13264.

B. V. Chirikov. A universal instability of many-dimensional oscillator systems. *Physics Reports*, 52(5):263–379, 1979.

C. Colas, O. Sigaud, and P.-Y. Oudeyer. How many random seeds? Statistical power analysis in deep reinforcement learning experiments. arXiv:1806.08295, 2018.

A. Daza, A. Wagemakers, B. Georgeot, D. Guéry-Odelin, and M. A. F. Sanjuán. Basin entropy: a new tool to analyze uncertainty in dynamical systems. *Scientific Reports*, 6:31416, 2016.

K. M. Deck, M. Payne, and M. J. Holman. First-order resonance overlap and the stability of close two-planet systems. *The Astrophysical Journal*, 774(2):129, 2013. arXiv:1307.8119.

D. W. Franklin, S. Franklin, and D. M. Wolpert. Fractionation of the visuomotor feedback response to directions of movement and perturbation. *Journal of Neuroscience*, 2019.

H. Furuta, T. Matsushima, T. Kozuno, Y. Matsuo, S. Levine, O. Nachum, and S. S. Gu. Policy information capacity: information-theoretic measure for task complexity in deep reinforcement learning. *International Conference on Machine Learning*, 2021.

C. Grebogi, S. W. McDonald, E. Ott, and J. A. Yorke. Final state sensitivity: an obstruction to predictability. *Physics Letters A*, 99(9):415–418, 1983.

D. Izzo and E. Öztürk. Real-time guidance for low-thrust transfers using deep neural networks. *Journal of Guidance, Control, and Dynamics*, 44(2), 2021.

L. Lai and S. J. Gershman. Human decision making balances reward maximization and policy compression. *PLOS Computational Biology*, 2024.

M. Matthews, M. Beukman, B. Ellis, M. Samvelyan, M. Jackson, S. Coward, and J. Foerster. Kinetix: investigating the training of general agents through open-ended physics-based control tasks. arXiv:2410.23208, 2024.

H. Lee, D. Hwang, D. Kim, H. Kim, J. J. Tai, K. Subramanian, P. R. Wurman, J. Choo, P. Stone, and T. Seno. SimbaV2: hyperspherical normalization for scalable deep reinforcement learning. arXiv:2502.15280, 2025.

B. H. Repp. Sensorimotor synchronization: a review of the tapping literature. *Psychonomic Bulletin & Review*, 12(6):969–992, 2005.

J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal policy optimization algorithms. arXiv:1707.06347, 2017.

A. Wang, T. Herbert, and S. Gao. On the fractal nature of the reinforcement learning objective landscape. *Advances in Neural Information Processing Systems*, 2023.

J. Wisdom. The resonance overlap criterion and the onset of stochastic behavior in the restricted three-body problem. *The Astronomical Journal*, 85:1122–1133, 1980.